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Review

Advancements in Mg-based scaffolds for bone tissue engineering: From design strategies to clinical perspectives

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Abstract

Mg-based scaffolds are emerging as architected, load-bearing constructs that couple temporary mechanical support with bioactive, self-resorbing behavior. This review integrates advances in geometric design, fabrication, and biological translation of Mg-based scaffolds. The study comprehensively discusses key design parameters, including porosity, pore size, pore shape, pore orientation, strut thickness, alloy composition, and hybrid surface modification strategies. It further highlights that triply periodic minimal surfaces and optimized lattices with tailored interconnected porosity achieve cancellous-bone-like mechanics while enhancing permeability, osteogenesis, angiogenesis, and pro-healing immunomodulation. Various fabrication routes including powder metallurgy, replica casting, laser powder-bed fusion and binder-jetting techniques are compared with respect to microstructural control, defect tolerance, corrosion-fatigue, and scalability. Surface modifications, including fluoride/Ca–P conversion layers, layered double hydroxides, and polymer-ceramic hybrid coatings, act as physical barrier and ion-regulating interfaces that slow the initial corrosion burst, buffer local alkalinity, and provide controlled Mg^{2+} release within a therapeutic window. Collectively, in-vitro and in-vivo investigations show that architected Mg scaffolds are biocompatible, generate only transient hydrogen, and degrade in vivo more slowly and uniformly than static immersion predicts. Bone ingrowth tracks the resorption front, enabling progressive load transfer from the scaffold to regenerating tissue. Compared with planar Mg devices, these lattices mitigate stress shielding, conform to irregular defects, and provide orders-of-magnitude greater surface for osseointegration. To address key translational challenges, including synchronizing scaffold degradation with bone healing, ensuring corrosion-fatigue reliability under physiological conditions, and meeting regulator-compliant manufacturing standards, the review highlights the role of Mg-based biocomposite scaffolds with hybrid surface treatments. These surface strategies improve structural persistence and biological compatibility. Together, they strengthen the pathway toward full-scale clinical translation of Mg scaffolds for orthopedic applications.

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Keywords: Mg-scaffolds; Bone regeneration; Scaffold geometry; Corrosion-fatigue; Orthopedic translation.

Abbreviations: **Mg**, Magnesium; **BTE**, Bone Tissue Engineering; **ECM**, Extracellular Matrix; **SBF**, Simulated Body Fluid; **PLA**, Polylactic Acid; **PLGA**, Poly(lactic-co-glycolic acid); **PCL**, Polycaprolactone; **TCP**, β-Tricalcium Phosphate; **HA**, Hydroxyapatite; **Mg^{2+}**, Magnesium Ion; **Ca–P**, Calcium–Phosphate; **LDH**, Layered Double Hydroxides; **TPMS**, Triply Periodic Minimal Surface; **BCC**, Body-Centered Cubic (lattice); **LPBF/SLM**, Laser Powder Bed Fusion/Selective Laser Melting; **AM**, Additive Manufacturing; **PM**, Powder Metallurgy; **DMEM**, Dulbecco's Modified Eagle Medium; **BMP-2**, Bone Morphogenetic Protein-2; **TGF-β1**, Transforming Growth Factor-β1; **VEGF**, Vascular Endothelial Growth Factor; **GMP**, Good Manufacturing Practice; **HTO**, Heat-Treated Oxide (coating); **MC3T3**, Mouse Calvarial Pre-Osteoblast Cell Line; **PDLLA**, Poly(D,L-lactic acid); **SF**, Silk Fibroin; **NaCl**, Sodium Chloride; **PBF**, Powder Bed Fusion (general); **SLM**, Selective Laser Melting (Mg AM process); **BJ**, Binder Jetting; **RE**, Rare Earth.

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1. Introduction

Mg-based scaffolds are emerging as architected, load-bearing advanced constructs coupling temporary mechanical support to damaged bone with bioactive, self-resorbing behavior. Bone defects, especially critical-sized defects exceeding the natural healing capacity, often >2–2.5 cm in length, represent a major clinical challenge [1]. Such defects may arise from trauma, infection, tumor resection, or congenital conditions and frequently result in non-union fractures if not managed appropriately [2–4]. Traditional solutions, such as autografts, allografts, and xenografts, have significant limitations, including donor site morbidity, limited supply, and the risk of immune rejection or disease transmission [5]. Autograft procedures carry complication rates around 8–10% and allografts can trigger immune responses [6]. Such drawbacks have spurred interest in bone tissue engineering (BTE) as an alternative approach to regenerate bone. BTE combines biomaterial scaffolds, osteogenic cells, and growth factors to facilitate new bone formation in significant defects. The scaffold serves as a temporary extracellular matrix, guiding cell attachment, proliferation, and differentiation while maintaining a space for the formation of new tissue. To be effective, a bone scaffold must be biocompatible, osteoconductive, ideally osteoinductive (able to induce stem cells to differentiate into bone-forming cells), and biodegradable at a controlled rate that matches the tissue healing process [7]. It should also possess suitable mechanical strength, with stiffness comparable to that of natural bone, to prevent stress shielding. Additionally, it should have an optimal porosity with interconnected pores that facilitate cell migration and vascular ingrowth, as well as surface properties that promote cell adhesion. Recent biomaterials research demonstrates the need for implants that couple mechanical competence with biologically active interfaces. Spark plasma-sintered Ti-Zn alloys show how microstructural control can enhance strength and corrosion resistance [8], while hydrogel and gelatin-heparin based scaffolds underscore the importance of biofunctional soft matrices in tissue regeneration [9,10]. First-principles studies on amino-acid adsorption onto Mg surfaces further reveal how interfacial chemistry governs degradation and early cell interactions [11].

Conventional scaffold materials include polymers, ceramics, and metals. Various synthetic and Natural Polymers, including polylactic acid (PLA), Poly(lactic-co-glycolic acid) (PLGA), and collagen, are often biocompatible and biodegradable but usually lack the mechanical strength required for load-bearing bone regions. Ceramics such as hydroxyapatite and β-TCP offer excellent osteoconductivity due to their similarity to bone mineral, yet they are brittle and may not degrade readily in vivo. Permanent metals, such as titanium or stainless steel, have high strength. Still, they are non-degradable and can cause stress shielding, due to a much higher elastic modulus than bone, as well as requiring a second surgery for removal. Biodegradable metals have emerged as a compelling alternative that could combine mechanical robustness with gradual resorption. Among these, magnesium (Mg) and its alloys have gained particular attention for bone scaffold applications due to several advantageous properties [12]. Mg is an essential nutrient in bone metabolism, and Mg^{2+} ions play key roles in osteoblast function and enzyme activation in the extracellular matrix. Mg alloys exhibit an elastic modulus of around 41–45 GPa, closer to that of natural cortical bone (10–30 GPa), potentially reducing stress shielding effects [13]. Moreover, Mg and its alloys are biocompatible and gradually corrode in physiological environments, eliminating the need for implant removal and creating space for new bone tissue as they degrade. The Mg biodegradability serves a dual purpose: the degrading scaffold provides a transient mechanical support and simultaneously releases Mg^{2+} ions that can stimulate bone regeneration by upregulating osteogenic factors, including TGF-β1 and BMP-2, as well as angiogenic factors (VEGF) in bone marrow stem cells, promoting new bone formation and vascularization [14].

Despite these advantages, pure Mg in the physiological environment corrodes too rapidly, releasing H_2 and OH^- ions as by-products of the corrosion reaction. If uncontrolled, rapid Mg degradation can lead to accumulation of hydrogen gas pockets at the implant site and a local increase in pH, both of which may impair bone healing and cause tissue damage. Contemporary research has addressed these limitations through alloying [15], surface treatments [16], heat-treatment-induced microstructural refinement [17,18], and severe plastic deformation approaches [19,20] to moderate Mg corrosion to a clinically tolerable rate. Alloying Mg with elements like calcium (Ca), zinc (Zn), manganese, and rare earths can improve corrosion resistance and strength, while coatings create a barrier to moderate the initial corrosion burst. Furthermore, combining Mg with biopolymers [21] and ceramics [22] in composite scaffolds offers a strategy to harness Mg's benefits while mitigating its drawbacks.

Mg scaffolds outperform conventional planar Mg plates in long bone defect repair by governing design-specific bio functionalities for clinical translation, as shown in Fig. 1. Solid Mg plates, despite Mg's relatively lower elastic modulus [23], can still out-stiffen healing bone, causing stress shielding, and their flat shape provides limited surface for bone on growth, leading to poor osseointegration [24,25]. Moreover, a planar plate cannot conform to irregular defect geometries, whereas a 3D-printed Mg scaffold can be patient-specific, precisely filling complex bone voids [26]. Planar Mg implants are also prone to accelerated, localized corrosion (pitting), which concentrates degradation and hydrogen gas release in specific areas, thereby undermining mechanical integrity and biocompatibility [27,28]. In contrast, porous Mg scaffolds with controlled architecture and interface treatments directly address these issues. The high-porosity, biomimetic lattice of a Mg scaffold mimics trabecular bone, tuning the implant's stiffness to closely match that of native bone and enabling deep tissue ingrowth for robust osteointegration [29]. It modulates Mg degradation from rapid and uneven to gradual and uniform, allowing the scaffold to maintain its strength as it resorbs in step with new bone formation [30]. This controlled

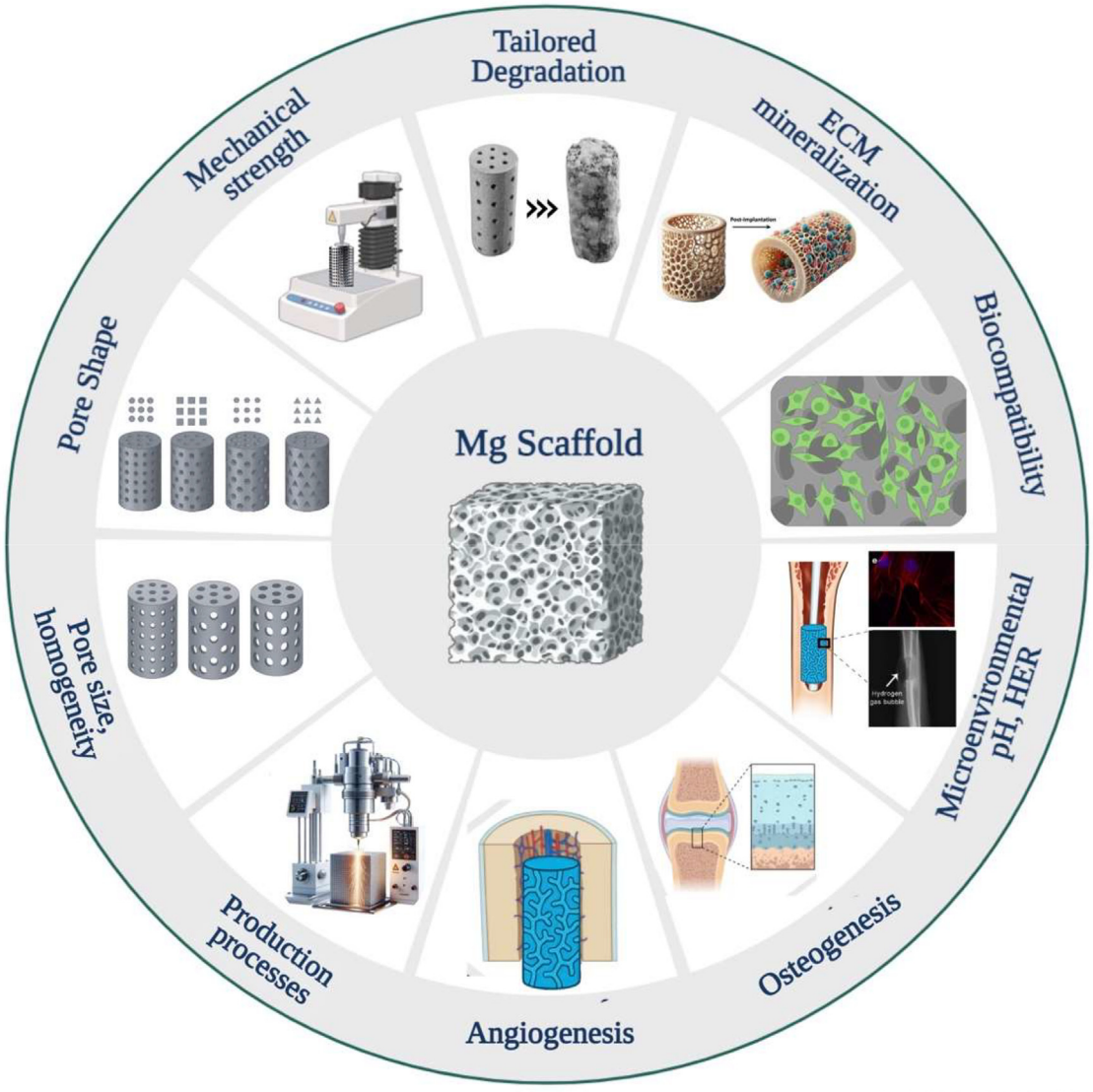


Fig. 1. Overview of the interlinked design and translational factors governing Mg-scaffold clinical application.

corrosion prevents early structural loss and avoids the local alkalinity seen with uncoated Mg, while the released Mg^{2+} ions and osteoconductive scaffold surfaces together stimulate osteogenic cell activity and angiogenesis, resulting in superior bone regeneration compared to planar implants [31].

Mg-based scaffolds have evolved significantly from simple fast-corroding implants to sophisticated engineered constructs. In this review, we outline the current state-of-the-art of Mg-based scaffolds for bone tissue engineering, discuss the various design aspects and fabrication strategies that have been employed to optimize their performance, examine the biological responses to Mg scaffolds in vitro and in vivo, and consider the remaining challenges for clinical translation of these scaffolds. Recent in vivo evidence demonstrating bone regeneration with Mg scaffolds, coupled with early clinical success of Mg alloy implants, underscores the potential of Mg-based scaffolds to revolutionize the treatment of critical bone defects.

Several review articles have previously examined Mg and Mg-alloy scaffolds [32,33], however, most have focused narrowly on either corrosion mechanisms, alloy design, or additive manufacturing [34–37]. In contrast, the present review provides a comprehensive and integrative framework that connects geometric design parameters including porosity, pore size and shape, unit-cell topology, and strut architecture with mechanical behavior, degradation kinetics, immunomodulatory responses, and in vivo bone regeneration. It further synthesizes evidence across metallic, composite, and surface-engineered Mg scaffolds to identify architecture-driven biological outcomes and translational requirements. By mapping geometry–property–biology linkages and aligning them with clinical design criteria, including corrosion-fatigue reliability, defect-conformal architectures, and regulatory considerations, this review uniquely addresses the practical question of how Mg scaffolds can progress from experimental constructs to clinically deployable, load-bearing implants. The rationale of

this work is therefore to consolidate these multidisciplinary insights into a unified, clinically relevant perspective that is currently missing from the existing literature.

2. Design aspects of Mg scaffolds for bone tissue engineering

Before detailing geometric design considerations, it is essential to contextualize the intrinsic material characteristics of Mg and its alloys. Mg possesses a low elastic modulus comparable to cancellous bone, favorable biodegradation driven by physiological corrosion, and inherent osteogenic and angiogenic ion release. Alloying with elements such as Zn, Ca, and rare-earth modifiers further tailors its corrosion resistance and mechanical stability, while maintaining biocompatibility. These material attributes form the foundation upon which architected scaffold geometries, such as TPMS, graded porosity, and optimized strut networks, are engineered to balance mechanical integrity, degradation kinetics, and biological performance.

Porosity and pore architecture are critical design factors dictating the performance of Mg scaffolds in load-bearing bone regeneration. By tailoring the percentage porosity, pore size, shape, and spatial pattern, the scaffold's mechanical integrity and degradation profile can be optimized, while promoting cell and tissue interactions. High porosity is generally desirable for facilitating cell/tissue ingrowth and vascularization, but it comes at the cost of reduced mechanical strength [38]. An optimal balance must be struck between porosity and structural stability. The influence of geometric variables on biological responses, including cell adhesion, osteogenesis, angiogenesis, and in vitro degradation behavior, such as corrosion rates and mechanical retention, is focused on in this section.

2.1. Porosity and pore size: trade-offs in mechanics and biology

2.1.1. Percentage porosity

Mg scaffolds are typically designed with ≥50% porosity to mimic the highly porous nature of cancellous bone and ensure space for cell colonization. However, increasing porosity dramatically decreases stiffness and strength following foam scaling laws [39]. An open-porous Mg scaffold with 55% porosity exhibits compressive strengths on the order of 40–46 MPa (cancellous bone range), whereas similar structures with 75% porosity dropped significantly [32].

The inverse porosity-strength relationship, risk of mechanical instability under load before new bone matures. Nonetheless, elevated porosity enhances osteoconductivity by exposing a larger surface area for cell attachment and facilitating tissue infiltration. Studies have noted that Mg scaffolds with ultrahigh porosity can still support bone defect healing if their architecture provides sufficient initial strength and if degradation is appropriately controlled [40,41]. In practice, graded porosity designs having higher porosity in the interior for bone ingrowth and lower porosity at load-bearing regions are being explored to reconcile these demands.

2.1.2. Pore size

The pore size must be large enough to permit cellular migration and neovascularization, yet small enough to maintain ample surface for cell anchorage. In vitro, smaller pores (50–100 μm) favor initial cell seeding density and surface attachment, whereas larger pores (>200 μm) better enable nutrient diffusion and tissue penetration as shown in Fig. 2. The porosity (70–80%) demonstrated in Fig. 2 results from the use of triply periodic minimal surface (TPMS) unit-cell architectures, where the porosity is mathematically defined by adjusting the solid volume fraction, together with fabrication routes such as laser powder-bed fusion and replica casting that allow fine control over strut thickness and pore size. For bone scaffolds, a minimum pore diameter of 100 μm is often cited, with pores >300 μm considered optimal for robust osteogenesis and angiogenesis [42].

In Mg scaffolds, Cheng et al. found that increasing an open pore size from 250 μm to 400 μm (at constant 55% porosity) significantly improved new bone formation in vivo by enabling more rapid early vascularization [44]. The larger pores facilitated deeper vessel invasion and nutrient supply, resulting in more mature bone tissue within the scaffold. However, this same increase in pore size can modestly accelerate in vitro corrosion: with fewer, larger pores, the structure experiences more direct fluid percolation, leading to faster Mg dissolution in immersion tests [32]. Thus, while bigger pores enhance biological performance by promoting angiogenic and osteogenic activity, they may require surface treatments to mitigate any accompanying increase in corrosion rate.

2.2. Pore shape and spatial architecture: effects on degradation and tissue ingrowth

2.2.1. Pore shape

The geometry of individual pores (e.g., cylindrical, spherical, cubic, gyroidal) influences both the stress distribution in the scaffold and how cells/interstitial fluids interact with the surface. Smooth, rounded pores (as in sphere-templated or gyroid scaffolds) tend to eliminate sharp corners that concentrate stress, thereby improving the scaffold's fatigue resistance under cyclic loads [45,46]. In contrast, angular pores or strut-based lattices (e.g., cubic lattice voids) may exhibit higher local stresses but also have a greater surface area. From a biological standpoint, pore shape affects how cells bridge and populate the scaffold interior. Curvilinear surfaces (e.g., gyroid minimal surfaces) present a continuous interface that cells can readily adhere to, potentially promoting more uniform cell colonization as shown in Fig. 3. In one comparative study, additively manufactured Mg scaffolds with identical porosity and pore size but different unit-cell shapes (diamond lattice, sheet-gyroid, and body-centered cubic) showed notable differences in behavior [47].

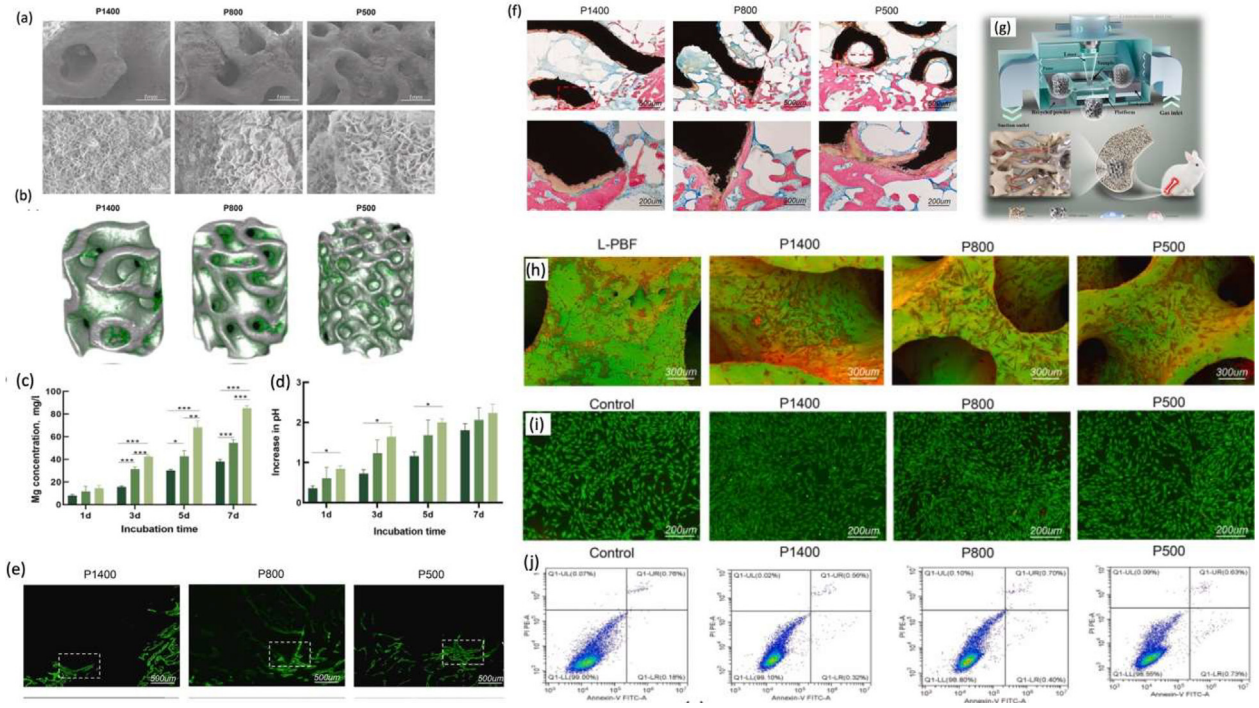


Fig. 2. Additively manufactured WE43 Mg scaffolds after high-temperature oxidation (HTO) are shown across three pore sizes (P1400, P800, P500); (a) SEM shows the surface morphology of each lattice; (b) 3D micro-CT reconstructions show corrosion products within the interconnected architecture. Extract studies demonstrate pore-size-dependent dissolution, with (c) Mg^{2+} concentration and (d) pH rise generally increasing as pore size decreases (P500 > P800 > P1400). In vivo, (e) fluorescent polychrome labeling and (f) methylene blue/acid fuchsin histology at 4 weeks show robust bone ingrowth at the bone–implant interface, with visible tissue penetration into pores. (g) Schematic depicts the laser powder-bed fusion (L-PBF) process and implantation in a rabbit femoral condyle defect. Cytocompatibility shown by (h) well-spread BMSC cytoskeleton on as-built L-PBF control and HTO lattices, (i) predominantly live-cell staining after 7 days, and (j) low apoptosis by Annexin V-FITC/PI flow cytometry across groups. Adapted with permission from [43].

All designs fulfilled minimum biocompatible criteria, but their mechanics and degradation varied: the gyroid and BCC lattices yielded higher compressive stiffness than a diamond-shaped lattice, and each exhibited distinct corrosion rates under identical testing. These pore-shape-driven differences extended to biological outcomes in vitro, including scaffolds with the gyroid and BCC architectures, which elicited higher osteogenic marker expression in mesenchymal stem cells than the diamond lattice [49]. Such results underscore that pore morphology itself can modulate the microenvironment for cells by altering surface topology and local ion release rates as Mg corrodes.

2.2.2. Spatial pattern (architecture)

Beyond individual pores, the spatial arrangement of pores, including ordered versus random, homogeneous versus gradient porosity, and oriented channels versus an isotropic network, plays a pivotal role in scaffold performance. Ordered lattices manufactured by 3D printing allow precise control of pore connectivity and orientation. This can be advantageous: aligning pore channels with the principal load axis can preserve higher mechanical strength at a given porosity [39], and it facilitates directed tissue ingrowth. A Mg-scaffold fabricated with uniaxially aligned cylindrical pores, using oriented wire-space holders, achieved mechanical properties within the range of human cancellous bone while maintaining open pathways for bone ingrowth along the pore axis [50]. Fig. 4 shows that oriented pores also influence degradation; fluid flow is less obstructed in straight channels, which can lead to more uniform but slightly faster corrosion internally. In contrast, a randomly porous foam may trap corrosion by-products that locally slow degradation [51]. Interconnectivity is another crucial aspect of spatial design. A highly interconnected pore network permits cell migration and vascular invasion throughout the scaffold; however, if connections are too narrow, they can become clogged with corrosion precipitates and tissue matrix. In biomimetic designs, hierarchical pore architectures, such as macropores for vessels and micropores for cell-matrix interactions, have been shown to improve outcomes [52,53]. For Mg scaffolds, incorporating small interconnecting windows (150–200 μm) between larger pores yields a network more akin to natural bone porosity, which has been associated with improved tissue infiltration and mechanical stability as the scaffold degrades [54].

The spatial heterogeneity of porosity can also be intentionally designed. A gradient structure with a dense surface and porous core could reduce initial burst corrosion by limiting fluid ingress at the surface while still providing a porous interior for bone growth [55]. Although most Mg scaffold studies employ uniform architectures for simplicity, emerging fabrication methods, such as extrusion-based printing and investment casting with space-holder gradients, are enabling

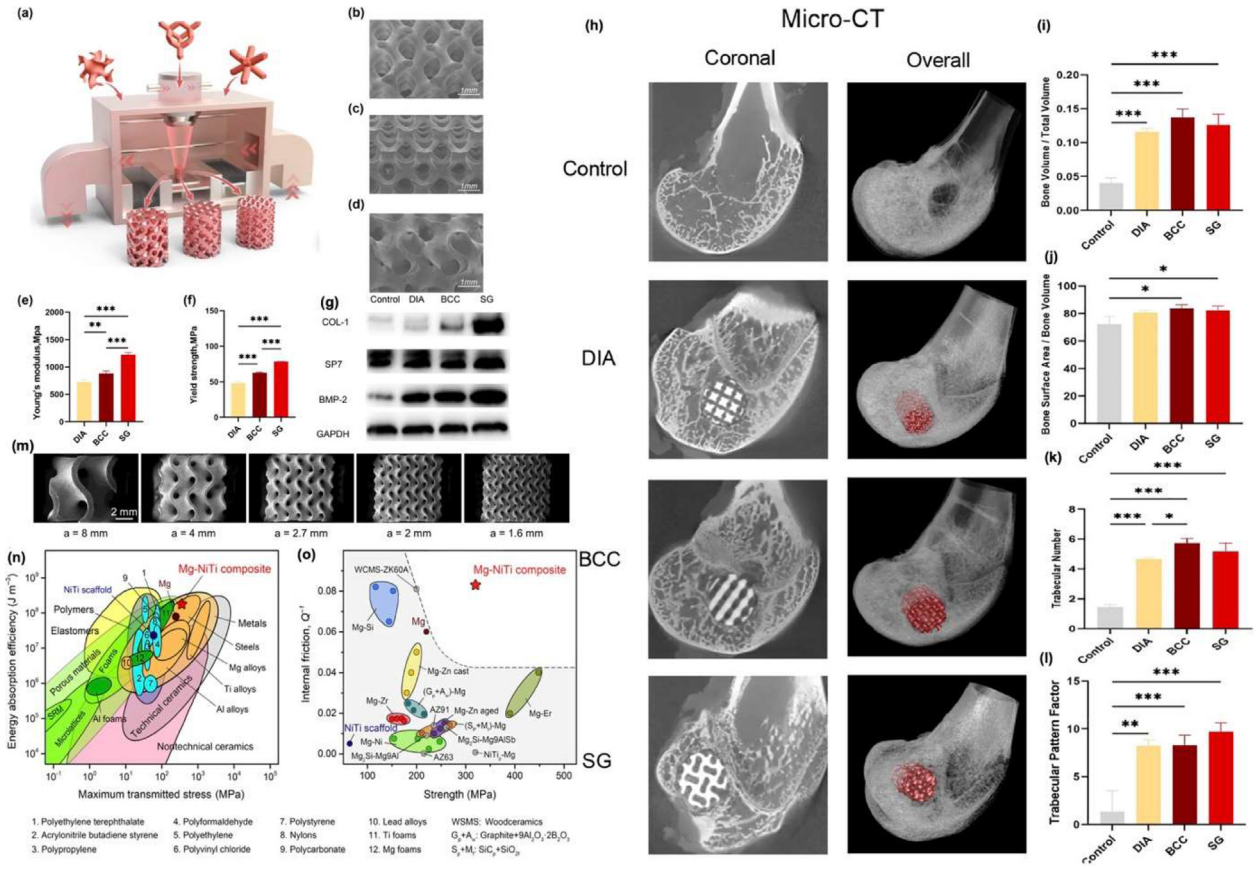


Fig. 3. (a) Laser powder-bed fusion (L-PBF) printing of three unit-cell architectures: diamond (DIA), body-centered cubic (BCC), and sheet-gyroid (SG), for Mg scaffolds. (b–d) SEM images show pore surfaces for DIA, BCC, and SG (scale bars: 1 mm). (e–f) Young's modulus and yield strength, showing architecture-dependent strengthening with SG > BCC > DIA. (g) Western blots of osteogenic proteins (COL-1, SP7/Osterix, BMP-2; GAPDH control) for BMSCs. (h) Micro-CT (coronal view and 3D rendering) of rabbit distal femoral-condyle defects implanted with DIA, BCC, or SG scaffolds versus empty control. (i–l) Quantitative micro-CT metrics demonstrating enhanced regeneration with architected Mg: bone volume fraction (BV/TV), bone surface area/bone volume (BS/BV), trabecular number (Tb.N), and trabecular pattern factor (Tb.Pf) (significance: $p < 0.05$; $p < 0.01$; $p < 0.001$). (m) Library of open-cell NiTi lattice preforms with decreasing unit-cell size used to create Mg–NiTi interpenetrating composites by pressure infiltration of molten Mg. (n) Ashby map of energy-absorption efficiency vs. maximum transmitted stress placing the Mg–NiTi composite. (o) Strength vs. internal friction chart highlighting the exceptional strength–damping synergy of the Mg–NiTi composite compared with Mg alloys, Al foams, and other structural materials. Adapted with permission from [45,48].

the fabrication of functionally graded Mg scaffolds. These architectures aim to deliver high surface integrity and strength to withstand surgical implantation and immediate loads, while maintaining high porosity internally to promote rapid osteointegration [56].

2.3. Geometric design influence on in vitro degradation behavior and biocompatibility

Mg's corrosion rate in physiological media is a crucial factor, as an overly rapid degradation can compromise mechanical support and produce harmful hydrogen gas pockets, causing tissue necrosis. Geometric design impacts corrosion in multiple ways:

2.3.1. Surface area-to-volume ratio

More porous scaffolds (and those with smaller pores) have a higher exposed surface area relative to bulk material. Initially, yields faster uniform corrosion as more Mg is in contact with the fluid. For instance, a highly porous Mg foam might exhibit a higher mass loss per unit time than a less porous one of the same alloy [57]. However, as mentioned, very fine pore structures can accumulate corrosion products ($Mg(OH)_2$, apatite) within small pores, which then act as a diffusional barrier and passivate further attack [58]. In contrast, scaffolds with large, open pores undergo continual fluid exchange, which washes away protective layers and maintains a higher corrosion rate. This was observed by Huo et al. [59] for Mg–Zn–Ca scaffolds, where nearly 500 μm porous specimens (75% porosity) exhibited rapid initial corrosion unless a protective coating was applied. In contrast, the same scaffold with a fluorapatite/MgF_2 coating showed significantly attenuated corrosion, preserving its structure for a longer period.

2.3.2. Pore connectivity and fluid flow

Scaffolds with highly interconnected pore networks tend to corrode more homogeneously. Fluid can access almost all

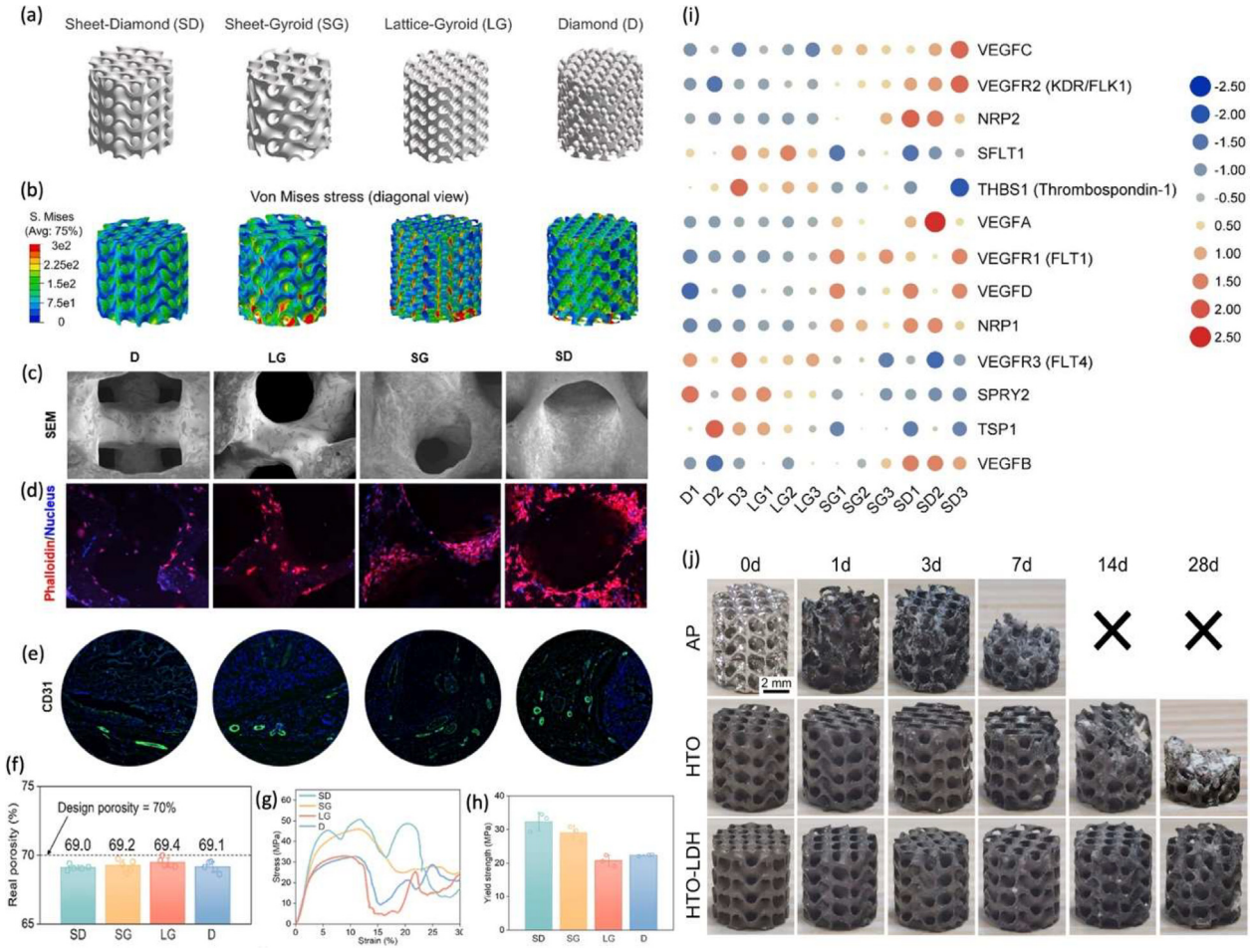


Fig. 4. (a) WE43 Mg Scaffold lattices: Sheet-Diamond (SD), Sheet-Gyroid (SG), Lattice-Gyroid (LG), Diamond (D), all designed at 70% porosity. (b) lower stress concentrations in sheet-type TPMS (SD/SG) and higher hotspots in strut-based LG/D. (c) SEM confirms that all four architectures have fully interconnected, smooth pores. (d) Phalloidin/nuclear staining evidences robust cell adhesion and spreading, most continuous on SD/SG surfaces. (e) CD31 immunofluorescence indicates denser microvessels around SD/SG implants. (f) Measured porosity closely tracks the design, with 70% porosity across groups. (g) Compressive curves highlight architecture-specific mechanics with broader plateau/energy absorption in SD/SG. (h) Yield strength ranks SD > SG > D > LG. (i) Angiogenesis gene dot-plot shows up-regulation of VEGFA/VEGFR2 (KDR), NRP1/2, and related markers in sheet-type lattices versus strut-based designs. (j) r-SBF immersion (0–28 d) compares surface states: as-printed (AP) deteriorates rapidly, high-temperature oxidation (HTO) slows attack, and HTO-LDH best preserves architecture with minimal pitting through 28 d Adapted with permission from [51].

internal surfaces, resulting in fairly uniform thinning of the struts [60]. Under cyclic loading in a flow environment, such open-porous Mg scaffolds have been shown to withstand corrosion without catastrophic failure of the network [61]. By contrast, designs with blind pores or occluded regions may develop localized corrosion hotspots (pitting) where fluid stagnates. Ensuring each pore is connected helps prevent the internal buildup of alkaline corrosion products that could accelerate localized material loss [62]. In-vitro studies using circulating media, open-porous Mg scaffolds maintained more stable pH profiles and more even degradation morphology than less-permeable designs [63]. Essentially, a well-ventilated architecture prevents "self-sealing" of pores by corrosion precipitates [64], but at the expense of a somewhat higher overall corrosion rate, again highlighting the need for corrosion-mitigating strategies (alloying or coatings) in highly open architectures.

2.3.3. *Structural redundancy and load transfer*

Mechanically, geometry dictates how the scaffold loses integrity as it corrodes. Designs with many redundant struts or continuous surfaces including trabecular-like lattices and gyroids, can tolerate a degree of material loss without abrupt failure [65]. They gradually shed load onto the forming bone. A Mg gyroid scaffold (60% porous) retained its load-bearing function longer during in vitro degradation compared to a stochastic foam of similar porosity [66], because the gyroid's continuous framework redistributed stresses even as walls thinned.

In contrast, a scaffold composed of discrete vertical pillars might fail more suddenly once critical cross-sections are corroded [67]. Researchers use in vitro mechanical testing after various immersion periods to evaluate this: acceptable designs show only a slow decline in compressive modulus over weeks, indicating that the scaffold can support forces

until substantial new bone is deposited [68]. The pore geometry should thus be optimized to degrade within biosafety limits, without early loss of mechanical integrity. Surface treatments on Mg scaffolds have exhibited more gradual mechanical decay, as corrosion first occurs superficially and corrosion products even temporarily reinforce the scaffold's struts [69]. Indeed, $Mg(OH)_2$ and apatite deposits can infill the scaffold's micro-pores and increase stiffness in the short term [70], a phenomenon noted to keep mechanical properties near the range of native bone for the initial weeks of degradation.

The geometric design of Mg scaffolds is a multivariate optimization challenge. Table 1 highlights representative studies illustrating how pore design parameters affect in vitro degradation and biological performance. These works collectively demonstrate that scaffold geometry must be engineered holistically to provide mechanical stability and controlled corrosion (for load support), while simultaneously creating a hospitable environment for cell attachment, vascular invasion, and bone formation. The convergence of additive manufacturing and biodegradable metallurgy now allows unprecedented control over these geometric parameters. As a result, today's Mg-based scaffolds can be architected to capitalize on Mg's favorable biomechanical and biological properties while mitigating its challenges (rapid corrosion). Ongoing research is extending these design principles, for example by exploring bio-inspired pore geometries with minimal surfaces and functionally graded porosity that further enhance the angio-osteogenic capacity of Mg scaffolds [71]. With optimized geometric design, Mg scaffolds are increasingly meeting the stringent requirements of load-bearing bone tissue engineering, combining adequate initial strength, osteopromotive degradation, and the ability to actively participate in the healing process by stimulating new bone and blood vessel formation.

3. Mg-based scaffold fabrication techniques

Engineering a Mg-based scaffold for bone tissue engineering requires not only the right material composition but also the proper structure. Achieving a highly porous, interconnected architecture in Mg (a relatively reactive and low-melting-point metal) is non-trivial, and a variety of fabrication techniques, shown in Fig. 5, have been developed and refined to create Mg scaffolds with controlled porosity and geometry. The significant categories of scaffold fabrication methods include powder metallurgy (sintering with space-holders), casting or foam templating, additive manufacturing (3D printing), and fiber-based techniques including electrospinning [77]. Fig. 5 illustrates some of the common fabrication strategies for Mg-based scaffolds, and each is discussed below along with its advantages and limitations.

3.1. Powder metallurgy and space-holder sintering

Powder metallurgical routes use Mg or Mg-alloy powders mixed with removable space-holders (e.g., salts, urea, polymers). After compaction and sintering below Mg's melting point, the space-holder is removed to form porosity. Studies using urea and HA as space-holders have produced Mg–Zn scaffolds with high porosity, strong compressive properties, and reduced corrosion rates [78]. Polymer-foam replication and salt-leaching approaches also generate highly interconnected networks [79], although the pore morphology may be irregular and the process control complex. Urea-assisted space-holder processing has also yielded Mg–Zn–HA scaffolds suitable for orthopedic repair [80].

3.2. Casting and foam-replication techniques

Replica casting involves infiltrating molten magnesium into polymer foams or sacrificial salt templates, followed by the removal of the template. These approaches can reproduce trabecular-like structures with interconnected pores. While simple and scalable, challenges include Mg reactivity, potential oxidation, and limited geometric precision.

3.3. Additive manufacturing (AM)

AM enables precise lattice geometries, controlled pore architectures, and patient-specific designs over conventional manufacturing techniques. Main AM techniques are described subsequently.

3.3.1. Material-extrusion printing

Material-extrusion (fused-deposition-type) systems extrude Mg-containing pastes or Mg-composite filaments into predefined lattice structures. Dong et al. [81] demonstrated Mg–Zn/bioceramic scaffolds with biomimetic unit cells. Although conceptually attractive, extrusion methods often produce coarse resolution and rough strut surfaces.

3.3.2. Laser powder-bed fusion (L-PBF/SLM)

L-PBF selectively melts Mg powders using a laser, enabling fabrication of porous lattices with highly dense struts. Liu et al. [82] achieved 99.5% dense WE43 struts with designed porosity, obtaining compressive strengths of 4–23 MPa and elastic moduli of 154–873 MPa, matching cancellous bone. Challenges include Mg oxidation, vaporization, and process instability.

3.3.3. Binder-jet additive manufacturing

Binder-jetting prints Mg powders using a liquid binder followed by sintering at sub-melting temperatures. Salehi et al. [74] fabricated Mg–Zn–Zr lattices with bone-like microporosity (13%), >95% pore connectivity, and high mechanical strength (tensile 130 MPa; compressive 349 MPa). This approach mitigates oxidation issues and improves geometric fidelity.

3.4. Fiber-Based and gas-foaming techniques

Fiber-oriented fabrication strategies (e.g., electrospinning, gas-foaming of polymer-Mg composites) create highly porous

Table 1
Influence of pore design parameters on scaffold degradation and biological performance.

Scaffold design	Porosity (%)	Pore pattern/Shape	In vitro degradation response	Biological response (in vitro/in vivo)	Ref.
Additively manufactured WE43 Mg scaffolds with three architectures (diamond lattice, sheet-gyroid, BCC)	60	Periodic lattice; pore size 500 μm (uniform)	HTO-treated scaffolds showed slow, uniform corrosion; distinct pore geometries exhibited different degradation rates (gyroid > BCC > diamond). All designs maintained structural integrity over 2 weeks of immersion.	All scaffold types biocompatible (no cytotoxicity). In vitro: Gyroid and BCC pores yielded higher ALP activity and mineralization vs. diamond. In vivo (rabbit femur): All scaffolds osseo-integrated with new bone; gyroid/BCC showed greater bone volume and more mature trabeculae formation (at 12 weeks) than diamond lattice.	[45]
LPBF-printed Mg scaffolds (WE43 alloy) with varied pore sizes and post-process oxidation	55–70 (adjusted via strut size)	Body-centered cubic lattice (regular strut network)	Without oxidation, higher surface area designs (smaller pores, 300 μm) corroded rapidly, losing 50% mass by 4 weeks. After oxide coating, corrosion rate dropped by 3 ×, and differences between pore sizes were minimized. Oxidized scaffolds retained 80% of initial compressive strength after 28 days in SBF.	In vitro: No adverse effect on cell viability (Mg^{2+} release remained in non-toxic range). Cells showed slightly greater proliferation on oxidized surfaces.	[72]
Powder-metallurgy Mg scaffolds with 60% porosity, two biomimetic designs (trabecular vs. oriented channel) +MgF_2 coating	60	Design A: Random foam-like (spherical 500 μm pores); Design B: Uniaxial channel pores (300 μm dia.)	Uncoated scaffolds showed fast initial degradation (pH rise > 9 within 48 h). MgF_2 coating significantly reduced the initial corrosion rate for both designs (50% lower hydrogen evolution) and delayed scaffold weight loss. Coating effectiveness was slightly better on the oriented-pore design (more uniform layer formation).	In vitro, both designs supported osteoblast adhesion; however, MgF_2-coated scaffolds exhibited higher cell viability. Coated channel-pore scaffolds induced earlier expression of osteogenic genes than coated foam scaffolds (attributed to guided cell alignment in channels). In vivo (rat femur): Coated scaffolds of both types healed critical defects with new bone, but oriented-pore scaffold showed more organized lamellar bone and higher vessel density in pores.	[73]
Vacuum-cast Mg-1.5Zn-0.2Ca scaffold high porosity and hierarchical pores	75	Spherical macropores (500 μm) with an interconnect 150 μm	As-cast scaffold (uncoated) exhibited rapid corrosion (complete loss of mechanical integrity by 2 weeks in DMEM). Applying MgF_2 + apatite coatings improved corrosion resistance dramatically: coated scaffolds corroded 3 × slower, maintaining structure > 4 weeks. The compressive yield strength remained roughly constant for 1 week in coated samples (compared to a drop in uncoated samples).	In vitro: Not reported in detail (study focused on degradation mechanics). Qualitatively, extracts from coated scaffolds supported proliferation of MC3T3 osteoblasts (indicating good cytocompatibility).	[59]
Binder-jet printed Mg scaffolds with hierarchical porosity	60–65	Interconnected lattice + micro-porous sintered necks	Binder-jet scaffolds exhibited relatively slow uniform mass loss; pH remained <8.2 due to reduced initial surface reactivity compared to LPBF Mg.	Supported osteoblast adhesion and proliferation; early mineral deposition observed by day 14.	[74]
Multiscale customized Mg scaffold (macro–meso–nano coupled design)	75	Gradient cylindrical pores with nano-texturing	Exhibited tempered corrosion with reduced hydrogen accumulation; local pH stabilized by nano-scale surface topography.	In vivo (rabbit radius) studies showed accelerated bone bridging and high expression of osteogenic markers.	[51]

(*continued on next page*)

Table 1 (*continued*)

Scaffold design	Porosity (%)	Pore pattern/Shape	In vitro degradation response	Biological response (in vitro/in vivo)	Ref.
Degradable Mg scaffold with MgF_2 + PDLLA hybrid coating	55–60	Cylindrical channels (400 μm)	The hybrid coating reduced the corrosion rate by 70% and suppressed early hydrogen release, while preserving mechanical integrity for over 4 weeks.	In vivo (rat femoral defect) showed significantly improved early-stage bone formation due to sustained Mg^{2+} release and moderated alkalinity.	[75]
Biodegradable Mg scaffolds with bioactive micro-arc oxidation (MAO) coating	60	Regular cubic lattice, 500 μm pores	MAO coating significantly reduced the initial corrosion burst and maintained pH <8.1; hydrogen generation decreased by 40%.	Enhanced osteoblast viability and mineralization; in vivo study showed faster cortical bridging and higher bone volume at 8 weeks.	[30]
Laser powder-bed fused WE43 scaffold with topology-optimized TPMS design	70–80	Schwarz-P and gyroid TPMS architectures	Topology-optimized TPMS reduced corrosion hotspots and produced uniform degradation fronts; pH rise stayed mild.	In vitro osteogenic differentiation significantly enhanced on gyroid vs. Schwarz-P. In vivo: TPMS scaffold demonstrated superior bone–implant interface maturity and vascular ingrowth at 8–12 weeks.	[76]

HTO, high-temperature oxidation (forming protective $MgO/Mg(OH)_2$ surface layer); LPBF, laser powder bed fusion.

composites where Mg particles or Mg precursors are embedded within biodegradable fibers. These structures are not load-bearing Mg scaffolds per se but provide favorable microenvironments for bone regeneration and can be integrated with Mg cores or lattices.

3.5. Hybrid and multi-step fabrication approaches

Many recent scaffold designs combine two or more fabrication techniques -e.g., AM-generated macro-architecture followed by salt-leaching to introduce micro-porosity, or L-PBF scaffolds subsequently coated or infiltrated with biomaterials. Such hybrid workflows optimize mechanical competency, degradation uniformity, and biological performance [80,84–88]. Table 2 summarizes the key features of such composite techniques utilized for Mg scaffold fabrication.

In practice, the fabrication method is chosen based on the specific application requirements. For load-bearing scenarios, such as weight-bearing extended bone defects, techniques that produce higher-density struts and predictable mechanical properties are preferred. These include 3D printing and carefully controlled sintering. For non-load-bearing or filler applications, such as cranial defects or as a bone graft extender, higher porosity foams or composite fiber scaffolds may be sufficient and offer better biological interactions. Importantly, post-fabrication spatial treatments often play a crucial role in final scaffold performance. Recent advances in polymer-based scaffold technologies further contextualize the role of Mg architectures in bone repair. Studies on 3D-printed polymer scaffolds have demonstrated improved print fidelity, tunable degradation, and enhanced osteoconduction [89],yet their limited load-bearing capacity underscores the need for mechanically robust metallic lattices. Emerging nanomaterial-enabled platforms designed for bone metastasis management highlight the importance of biofunctional cues, aligning with Mg scaffolds' intrinsic ion-mediated osteogenic and angiogenic potential [90]. Likewise, high-temperature printed PEEK implants exhibiting superior mechanical performance emphasize the growing demand for structurally resilient, defect-conformal implants [91].

4. Current state of Mg-based scaffolds

Mg alloy-based scaffolds encompass a broad range of material formulations and architectures, from monolithic porous Mg/Mg alloy structures to hybrid composites containing Mg in combination with polymers and ceramics. Research over the past decade has demonstrated rapid progress in tailoring the properties of these scaffolds to meet the stringent requirements of bone regeneration. Table 3 provides a summary of representative recent studies on Mg-based scaffolds and their key outcomes, reflecting the diversification in scaffold designs.

One primary class of materials in this domain is Mg alloys combined with one or more alloying elements to improve their mechanical and corrosion behavior as shown in Fig. 6. For load-bearing bone scaffold applications, alloys such as Mg–Zn, Mg–Ca, Mg–strontium (Sr), and rare-earth-containing alloys (e.g., WE43, which contains Y and Nd) have been explored [99]. Alloying can significantly slow the degradation; for example, a porous scaffold made of Mg–4%Zn–0.2%Ca (a ternary alloy, ZK40 series) showed a more gradual corrosion and longer integrity in vivo compared to pure Mg [100]. However, one must carefully choose alloy elements for biomedical use to avoid toxicity; Aluminum, is present in some commercial Mg alloys but is generally avoided in biodegradable implants due to concerns about chronic toxicity [101]. Recent research favors alloying with elements that

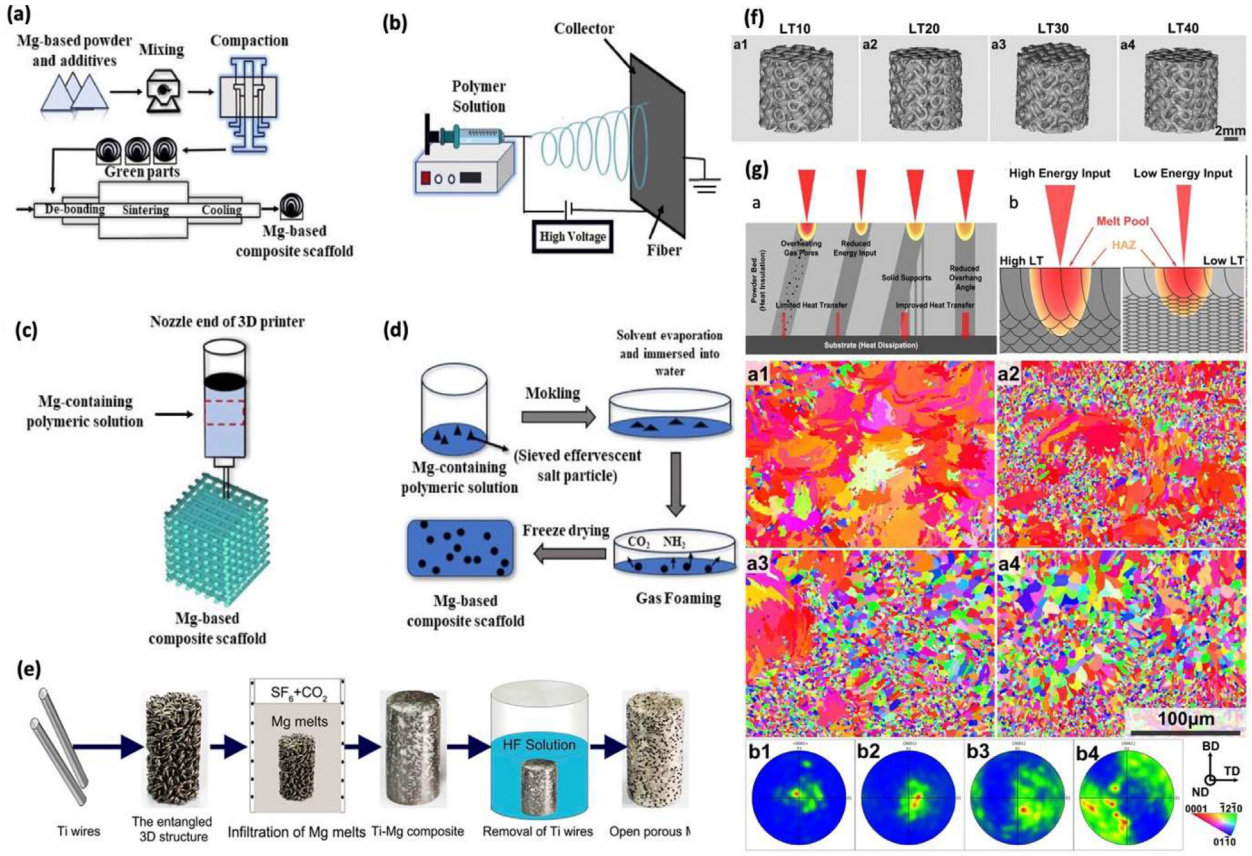


Fig. 5. (a) Powder-metallurgy steps for fabrication of Mg-based scaffolds. (b) Electrospinning of Mg-containing polymer solutions produces nanofibrous mats. (c) Nozzle-based 3D printing of Mg-loaded inks builds designed lattices at low temperature. (d) Effervescent-salt gas-foaming generates interconnected pores via in-situ CO_2/NH_3 release. (e) Ti-wire templating with Mg melt infiltration and selective Ti removal creates open-porous metallic Mg with fully interconnected channels; (f.a1–f.a4) L-PBF sheet-gyroid lattices printed at LT10, LT20, LT30, LT40 show progressively improved overhang fidelity and feature resolution as exposure/heat-sink conditions are tuned. (g.a) represents that excess energy promotes overheating and gas porosity; adequate heat dissipation supports the stabilization of melt pools. (g.b) high-LT creates deeper pool & larger HAZ; low-LT results in a shallower pool with reduced HAZ. (g.a1–g.a4) EBSD IPF-Z maps (LT10 to LT40, 100 μm) show a transition from mixed columnar–equiaxed bands toward more uniform, finer equiaxed grains at optimized LT (LT20–LT30), with fewer melt-pool bands (g.b1–g.b4) {0001} pole figures reveal a basal fiber texture ⟨0001⟩ is strongest at high-energy LT and weakens at intermediate LT, implying reduced anisotropy and more isotropic mechanical response. Adapted with permission from [83,36,44].

are either essential minerals, including Zn, Ca, and Sr [102] and have known biocompatibility in trace amounts like yttrium [103]. These alloying additions not only slow corrosion but also often refine the microstructure and increase strength. Adding rare earth elements can improve the yield strength and creep resistance of Mg, making the scaffold more capable of withstanding physiological loads [104]. A notable example is the MAGNEZIX® alloy (Mg–Y–RE–Zr) screws that have been used clinically in orthopedic surgery, demonstrating that Mg alloys can achieve sufficient strength and acceptable corrosion behavior for real-world use [105]. The success of these screws (used for fixation in foot and ankle surgeries, among others), which showed equivalent outcomes to titanium screws in short-term follow-up [106], provides confidence in the feasibility of Mg alloys for bone applications, though scaffolds for large bone defects impose additional design challenges.

Mg-based composite scaffolds, which integrate Mg with polymeric or ceramic phases. The rationale is to capitalize on the favorable attributes of multiple materials: Mg provides mechanical support and stimulates bone cell activity via released ions, while polymers or ceramics can enhance the scaffold's handling, bioactivity, or control of degradation. One approach has been to coat or fill a Mg scaffold with bioactive ceramics (like calcium phosphates) to slow corrosion and promote osteointegration. A 3D-printed Mg scaffold coated with a thin layer of octacalcium phosphate showed notably reduced initial degradation and improved bone formation in vivo compared to an uncoated Mg scaffold [107]. Similarly, Mg–HA (hydroxyapatite) composites have been explored. Incorporating HA particles into a porous Mg matrix can buffer the local pH as Mg corrodes and also act as nucleation sites for bone mineral, thus potentially enhancing bone ingrowth [108]. On the polymer side, biodegradable polymers like polycaprolactone (PCL), poly-L-lactic acid (PLLA), and gelatin have been combined with Mg in various ways [109]. One strategy is to incorporate Mg micro/nanoparticles into polymer scaffolds to create composites that slowly release Mg^{2+}. A recent study added nano-sized MgO (magnesium oxide) into a PCL/β-TCP scaffold and found that the Mg-containing scaffold signifi-

Table 2
Standard fabrication techniques for Mg-based scaffolds and their characteristics.

Fabrication method	Porosity & geometry control	Advantages	Limitations
Powder metallurgy + space-holder sintering	Reasonable control over total porosity (%) and pore size distribution (via space-holder particles), though pore shape is typically irregular (following particle shapes).	Simple equipment; can uniformly alloy or add ceramic phases; suitable for medium porosity (30–70%).	Risk of incomplete sintering or oxidation; mechanical strength of necks can be low; not ideal for huge pores (>1 mm) unless large particles are used.
Casting and foam replica	Porosity can be high (>70%); the pore shape mimics the template (e.g., polymer foam yields a reticulated structure). Macro-porosity control is moderate; micro-pores may form in walls.	Can produce large scaffolds; fully metallic struts (dense Mg) yield better strength; suitable for high-porosity foams.	Requires handling molten Mg (safety concerns); pore architecture depends on template quality; may have thicker struts and very open pores (less surface area).
Additive manufacturing (SLM, 3D printing)	Excellent control: CAD-designed pore size, shape, and gradient; complex and patient-specific geometries possible; consistent interconnected porosity.	Precise and reproducible architecture; enables high complexity (e.g., lattices, gyroid pores); potential for patient-specific implants.	Expensive and technically challenging (Mg powder flammability, equipment cost); printed parts may have internal residual stresses or require post-processing (heat treatment); surface may need further smoothing/coating.
Electrospinning (Mg-loaded fibers)	Produces fiber networks with nano-/microscale porosity between fibers; not a traditional "bulk" porosity, but can layer fibers for 3D thickness. Alignment of fibers can be controlled to some degree.	Very high surface area; easy to incorporate Mg compounds and other biomolecules; fibers can mimic extracellular matrix.	Not load-bearing by itself (used as membranes or filler); limited thickness (usually <1 mm per sheet without losing structure); Mg content must be in particle form (metallic Mg particles might be reactive during spinning).
Machining/Drilling	Porosity is low, and geometry is simple (e.g. straight drilled channels). Not a primary method for high porosity, but it can create uniform cylindrical pores.	Utilizes bulk alloy with known properties, achieving precise hole placement and size through CNC drilling.	Very low achievable porosity (often <30%); limited to simple geometries; significant material waste.

cantly outperformed the Mg-free scaffold in a rat calvarial defect, with greater new bone volume and more robust angiogenesis [110]. The Mg oxide not only provided a reservoir of Mg^{2+} ions to stimulate bone cells but also improved the mechanical stiffness of the polymer/ceramic scaffold, illustrating the synergistic effect [111]. Another approach is surface functionalization of polymer scaffolds with Mg, coating polymeric 3D scaffolds with a thin Mg layer or integrating Mg-alloy fibers within a polymer matrix, yielding a hybrid scaffold that benefits from both materials [112]. In summary, current Mg-based scaffolds can be roughly categorized as: (a) porous Mg/Mg-alloy scaffolds (bulk metallic structures, often with coatings), (b) Mg–ceramic composites (e.g., Mg with HA or bioactive glasses), and (c) Mg–polymer composites (either Mg particles in polymers or polymer-coated Mg frameworks). Each category addresses the core challenge of balancing mechanical support, corrosion rate, and bioactivity as shown in Fig. 7.

5. In vivo performance of Mg-based bone scaffolds

From a structural perspective, precisely tailored pore geometries and gradients lead towards biomimetic designs. Particularly, scaffolds with varying pore sizes and distributions have been widely reported for mimicking cancellous bone to maximize osteoconductivity [114]. The concept of hierarchical structuring has also emerged: scaffolds are being designed with multi-scale features, including macro-scale shape matching the defect geometry [115], micro-scale porous network for cell ingrowth [116], nano-scale surface textures or coatings to guide cell behavior [117]. Table 4 summarizes the in vivo performance of various Mg-based bone scaffolds. A recent breakthrough study created a Mg scaffold with such hierarchical architecture, including a trabecular-like triply periodic minimal surface (TPMS) pore design and a nano-layered surface coating, resulted in markedly improved bone regeneration in vivo compared to traditional uniform-pore scaffolds [118] indicating the significance of rationally designing porosity and surface functionalization to optimize biological performance.

In vivo studies show that moderate Mg^{2+} release (below 10 mm) stimulates osteoprogenitor cells, upregulating ALP, RUNX2, collagen, and angiogenic factors, including VEGF, to create a pro-regenerative niche. In contrast, rapid corrosion (high pH, >10 mm Mg^{2+}) can be toxic [119]. Importantly, animal experiments have confirmed excellent biocompatibility and safety of porous Mg scaffolds. Liu et al. [120] found that rabbits implanted with 3D porous Mg scaffolds had no significant rise in serum Mg or organ enzymes (ALT, BUN/Cr) and normal histology of liver, kidney, heart, and spleen throughout 3 months. Early X-rays revealed small radiolucent "bub-

bles" around the implant at 2 weeks, indicating hydrogen gas from corrosion. However, these gas pockets quickly diffused away and did not hinder healing. In fact, no chronic inflammation or fibrous encapsulation was observed: adjacent bone remodeled normally with healthy osteoblasts lining the degrading Mg surface [121]. In-vivo Mg scaffolds cause at most mild acute inflammation and transient gas release [122]; the body buffers the byproducts by renal excretion of Mg^{2+} and diffusion of H_2, so that systemic toxicity is negligible [123].

Table 3
Comprehensive analysis of Mg-based scaffolds based on formulation material.

Studies reporting structural mechanical and degradation behavior of architected Mg alloy scaffolds.

Composition	Scaffold geometry/Parameters	Compressive strength	Density/Porosity	Key performance	Ref.
Mg–1.5Zn–0.2Ca alloy	Spherical pore structure (450–600 μm pores, 75% porosity)	4.0 MPa (compressive yield)	0.75 (75% porosity)	MgF_2 surface coatings improved corrosion resistance. Sufficient strength for load-bearing; proposed for bone tissue engineering.	[59]
Pure Mg (3D-printed, hierarchical)	Patient-specific TPMS/hierarchical design (macro–meso–nano scales)	48.8 MPa (compressive)	0.30 (70% porosity)	Dual coatings (oxidized film + hydrotalcite) "tempered" degradation. Showed seamless bone integration and superior in vitro/in vivo regeneration.	[51]
WE43 Mg (RE alloy)	L-PBF additive lattice (cylindrical, diamond cells); struts ∼300–500 μm	4.4–23.5 MPa (comp., var. by design)	(50% porosity)	Mechanical properties comparable to cancellous bone (E 0.15–0.87 GPa). Good biocompatibility in vitro/in vivo, though rapid degradation led to loss of integrity by 4 weeks.	[82]
ZE21 Mg alloy	Machined lattice (wire-EDM), ∼60% porosity	–	0.40 (40% porosity)	Fluoride-passivated + PDLLA–BMP-2 composite coating slowed corrosion and released growth factor. In a rat femur defect, BMP-2–loaded scaffold induced bone formation along the pore network.	[75]
Mg + LIPUS	Regular porous scaffold (implanted in rat femur)	–	–	Low-intensity pulsed ultrasound (LIPUS) accelerated scaffold degradation but *enhanced* mineral deposition. At 24 weeks, new bone hardness with LIPUS was 15% higher, approaching healthy bone.	[92]

Studies on Mg–ceramic and Mg–polymer composites enhancing strength, stability, and bioactivity.

Composition	Scaffold geometry/Parameters	Compressive strength	Density/Porosity	Key performance	Ref.
Mg–4%Zn + nHA (5%)	Powder metallurgy + space-holder (urea)		0.60 (40% porosity)	Adding 5% nano-hydroxyapatite raised strength and stiffness vs. Mg–4Zn alone, while 7.5% HA caused a drop. 5% HA scaffolds showed slowest corrosion and best in vitro bioactivity.	[78]
Mg–NiTi interpenetrating	3D-printed Nitinol scaffold infiltrated with Mg	518.7 MPa (compressive)	1 (fully dense)	Achieved exceptional strength and ductility (35% strain). Shape-memory effect enables ∼60% recovery of 10% strain on heating, providing unprecedented damping and energy absorption.	[48]
Mg-alloy rods + bioglass cement	"Ferroconcrete" design: Mg-alloy core rods (2 mm) in osteoconductive bioglass–$Mg_3(PO_4)_2$ cement	180 MPa initial; 100 MPa at 6 wks	cortical-like density	Combines high strength and controlled biodegradation of Mg-alloy with bioactive cement. Matched cortical bone strength; gradually degraded and was replaced by mature new bone in rabbits.	[93]
Mg + β-TCP (15%vol.)	DIW 3D-printed β-TCP preform, liquid Mg infiltrated (continuous MMC)	Increased by 27% compared to pure Mg	1 (highly densified)	In situ reaction during infiltration formed Mg_2Ca, CaO, MgO phases. Strong interfacial bonding and phase dispersion yielded ∼27% higher compressive and 15% higher yield strength than pure Mg.	[94]
Mg foam + graphene (GNP)	Cast open-cell Mg foam, GNPs added (0–0.3 wt%)	optimum at 0.1% GNP	0.30–0.40 (foams)	Small graphene additions increased the compression plateau stress and energy absorption capacity, but higher GNP content caused a drop. Optimal GNP-reinforced foams showed enhanced stiffness and damping without embrittlement (study data).	[95]

Table 3
(*Continued*).

Studies demonstrating corrosion control and enhanced biofunctionality through coating strategies.

Base Mg scaffold	Coating (type)	Density/Porosity	Key performance	Ref.
Extruded Mg scaffold	MgF_2 & Ca–P (dip-coated)	55% porosity	Porous AM scaffold (55% void) remained mechanically within cancellous bone range even after immersion. Fluoride + CaP duplex coating slowed biodegradation to 0.2 mm/yr and improved biocompatibility.	[96]
LPBF Mg–Nd–Zn–Zr (JDBM)	Sr-doped octacalcium phosphate	0.50 (TPMS lattice)	Biomimetic TPMS scaffold promoted vascular in-growth. Sr-OCP coating increased surface roughness and hydrophilicity, enhancing cell adhesion and osteogenic differentiation. Coating significantly reduced Mg scaffold's degradation while releasing $Mg^{2+}/Zn^{2+}/Sr^{2+}/Ca^{2+}$, leading to superior in vivo bone regeneration.	[30]
3D-printed Mg–Nd–Zn–Zr	ZA-loaded Ca–P ceramic (dip-coated)	0.60 (lattice scaffold)	Zoledronic acid (bisphosphonate) in a ceramic coating on Mg alloy scaffold significantly slowed its degradation and provided sustained drug release. In osteoporotic rat femur defects, the drug-coated Mg scaffolds showed improved healing, with greater new bone formation and inhibited osteoclast activity compared to uncoated scaffolds.	[97]
Pure Mg (micro-arc oxidized)	Oxidized dextran + nHA "self-healing" polymer	0.50 (foam lattice)	A MAO-derived Mg–HA layer coated with oxidized dextran endowed the Mg scaffold with self-healing anticorrosion capability. The coated scaffold demonstrated antibacterial and antioxidant effects, significantly enhancing osteogenesis while reducing cell apoptosis in infected bone defects.	[98]
Mg–Zn–Ca alloy (ZE21)	MgF_2 + PDLLA (BMP-2 loaded)	0.55 (passivated lattice)	Fluoride coating plus a PDLLA film for rhBMP-2 delivery controlled the scaffold's degradation and released growth factor over time. In vivo, the BMP-2–coated Mg scaffold induced substantial new bone formation along its pores, outperforming the scaffold without BMP-2.	[75]

(RE, rare-earth; TPMS, triply periodic minimal surface; L-PBF, laser powder bed fusion; LIPUS, low-intensity pulsed ultrasound.) Compressive strengths are given for dry scaffolds unless noted. Yield strengths are listed when separately reported. Porosity is given as fraction of total volume.

5.1. Enhanced bone regeneration and integration

Surface-directed nanoscale modifications of the scaffold trigger mechanism-level shifts in vivo, prolonging structural persistence while reshaping hydrogen-evolution and tissue interactions. Posada et al. [124] demonstrates that Directed Plasma Nanosynthesis (DPNS) induces a mechanistic switch in AZ31 diamond-lattice Mg scaffolds by forming a nanostructured $MgCO_3$/MgO–$Mg(OH)_2$ surface stack (1–2 nm carbonate over 4.7 nm oxide/hydroxide), enriching surface Mg (from 1.5 at% to 14.5 at%) and reducing adventitious carbon (from 75.0 at% to 32.1 at%). XPS–EI correlations, as shown in Fig. 8, indicate earlier passivation and a more positive corrosion potential at 14 days, with comparable I_{corr}, consistent with the evolution of a multilayer film. In vitro immersion (DMEM) reveals front-loaded Mg release followed by slowdown, while in vivo DPNS implants retain structural integrity to 8 weeks versus complete resorption of controls by 4 weeks. μCT quantifies larger retained struts (505 ± 95 μm at 8 weeks) and distinct H_2 dynamics as single, larger gas pockets (upto14 mm) for DPNS versus multiple smaller bubbles (upto 5 mm) in controls. Collectively, DPNS tunes degradation from periphery-driven loss to core-moderated, more uniform corrosion, enhancing scaffold persistence but introducing a pore-clogging risk from denser corrosion products that must be balanced for biomedical use. Bonithon et al. [61] investigated how MgF_2-coated, open-porous Mg scaffolds evolve under dynamic immersion, capturing both the spatial pattern of attack and pore-scale morphometry over 2–14 days. n the longitudinal μCT reconstructions (Fig. 8k), red-marked voxels pinpoint corrosion that advances with time yet remains spatially confined, predominantly at the periphery, so the fiber network and overall cylindrical geometry persist across 2–14 days. The pore-diameter color maps show only modest architectural drift: while some outer pores enlarge, the scaffold exhibits a net shift toward smaller internal pores, consistent with debris-driven densification. Quantitatively, this manifests as a significant rise in volume fraction ($\Delta\varphi = +1.9\%$ at day 2 to $+9.7\%$ at day 14) alongside a progressive reduction in mean pore diameter (-3.6% at day 2 to -14.9% at day 14). Together with the material-loss maps, which localize damage rather than indicating wholesale strut failure, these results demonstrate that the fluoride-coated, entangled Mg scaffold withstands corrosion while retaining its architecture and limiting net volumetric loss, thereby sustaining scaffold integrity.

5.1.1. High bone volume & density

In osteoporotic rat defects, porous Mg-Zn alloy scaffolds restored bone volume and quality more effectively than controls. Zhu et al. [125] showed nearly complete defect repair and higher bone mineral density with Mg scaffolds. Mechanistically, Mg^{2+} was found to activate Wnt/β-catenin and bone morphogenetic pathways (Wnt5a, β-catenin, BMP-2), driving osteogenesis even in osteoporotic bone.

Table 4
Comparative in vivo performance of magnesium-based bone scaffolds conducted by various research groups.

Study (Year)	Mg scaffold	Animal model (Defect)	Key outcomes	Ref.
Duan et al., 2020	Mg-alloy rods + bioglass cement composite scaffold	Rabbit radius (critical-sized cortical defect)	Achieved cortical bone-matching strength (100 MPa after 6 weeks) and gradually degraded in vivo, being replaced by mature new bone with high osteoconductivity.	[93]
Dong et al., 2020	3D-printed porous pure-Mg scaffold (solvent-cast printing)	Rat femur (segmental defect) *(assumed)*	Interconnected porous Mg scaffold exhibited controlled biodegradation and good osteoconductivity; in vivo implantation showed effective bone regeneration in the defect site (complete bridging of the bone gap).	[130]
Bonithon et al., 2021	Sintered open-porous Mg scaffold (~55% porosity)	*No implant – ex vivo mechanical study*	Porous Mg scaffolds displayed fully interconnected pores and bone-like mechanical behavior. They withstood strains beyond daily physiological levels without failure, indicating potential to support load-bearing defects.	[131]
Zhang et al., 2022	3D gel-printed Mg scaffold with DCPD (Ca–P) coating	Rat femur (critical-sized defect)	DCPD-coated Mg scaffolds had slower corrosion and good biocompatibility. No systemic magnesium toxicity was observed, and micro-CT revealed abundant new bone formation at 6 weeks. At 12 weeks, 52% of the scaffold remained, with ongoing osteogenesis observed up to 24 weeks.	[132]
Wang et al., 2023	Additively-manufactured Mg scaffold (oxidized surface)	Rabbit femur (segmental defect) *(assumed)*	High-temperature oxidation treatment improved the scaffold's corrosion resistance while maintaining porosity. In vivo, oxidized Mg scaffolds provided stable mechanical support and enhanced new bone formation; an optimal pore size balanced strength and osteointegration, leading to robust defect healing.	[43]
Zhu et al., 2023	Biomimetic porous Mg–Zn alloy scaffold	Rat femur (osteoporotic bone defect model)	Mg alloy scaffold significantly promoted repair of osteoporotic bone defects. Treated defects showed greater bone mass and higher expression of osteogenic markers (e.g. Wnt5a, β-catenin, BMP-2) compared to controls, indicating activation of Wnt/β-catenin signaling for bone regeneration.	[125]
Mao et al., 2024	Silk-fibroin/MgO composite scaffold (shape-memory)	Rat skull (irregular cranial defect)	A water-activated shape-memory SF/MgO scaffold conformed to irregular defects and maintained mechanical stability during degradation. It enhanced osteoblast adhesion/proliferation in vitro and significantly improved in vivo bone regeneration and angiogenesis in skull defects compared to non-Mg controls.	[76]
Wang et al., 2024	Mg-enriched bovine bone matrix (ABBM-Mg) scaffold	Rat mandible (segmental defect)	The Mg-modified bone matrix scaffold demonstrated improved osteogenesis in mandibular defects. In vivo tests demonstrated faster bone defect bridging and higher bone density with the Mg-enriched scaffold compared to the unmodified bone matrix, without any adverse effects. This suggests Mg incorporation enhances the osteoconductivity of allograft bone scaffolds.	[133]
Lv et al., 2025	Multi-scale hierarchical Mg scaffold (dual-layer coated)	Rabbit femur (custom irregular defect)	A hierarchically structured Mg scaffold (macro–nano scale features) with a dual coating (oxide + hydrotalcite) demonstrated superior load-bearing capacity and tempered degradation. In vivo, it outperformed traditional scaffolds by achieving faster and more complete regeneration of irregular bone defects, marking a breakthrough in personalized bone repair.	[51]
Ye et al., 2025	Sr-enriched Mg alloy scaffold (SrOCP biocoating)	Rat femur (critical-sized condyle defect)	Strontium–OCP-coated Mg scaffolds (JDBM/SrOCP) markedly enhanced vascularized bone regeneration. In vivo results showed greater new bone volume and density, with upregulated osteogenic markers (ALP, COL-1, OCN) and angiogenic factors (CD31, VEGF) in the Sr-coated group vs. uncoated Mg.	[30]

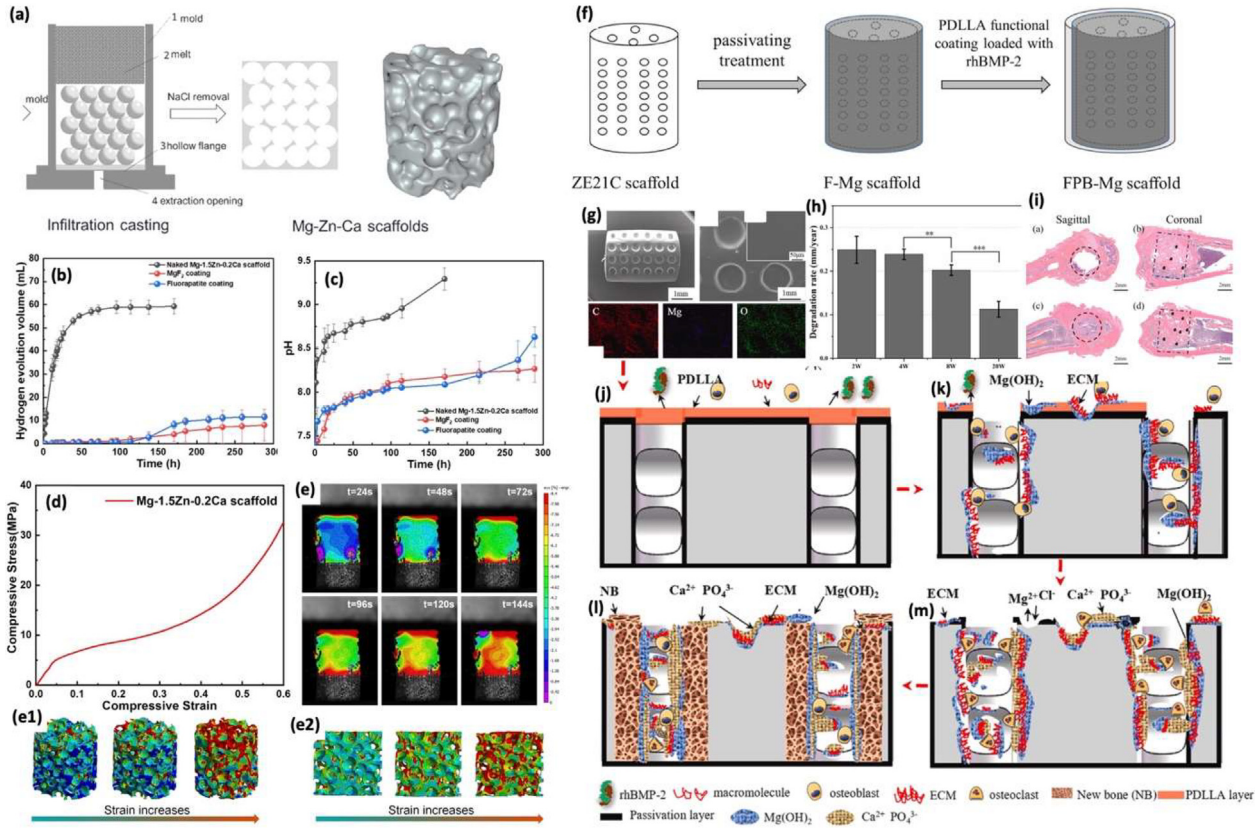


Fig. 6. (a) Vacuum infiltration casting of spherical-pore Mg–1.5Zn–0.2Ca scaffolds using a NaCl template yields an open, interconnected architecture. (b–c) Lower H_2 evolution and moderated pH rise for MgF_2 coated versus uncoated controls. (d) Compressive stress–strain displays the characteristic elastic–plateau–densification response with a low-density lattice. (e) Digital image correlation (DIC) maps reveal strain localization at ends/walls during loading; (e1–e2) FE simulations reproduce the evolving hot-spot bands as strain increases. (f) ZE21C porous scaffolds treated with PDLLA coating loaded with rhBMP-2 (g) SEM/EDX and fluorescence confirm uniform passivation and surface chemistry. (h) Longitudinal immersion shows reduced degradation depth for F-Mg, with further suppression for FPB-Mg. (i) Histology demonstrates enhanced bone fill along pores in FPB-Mg versus controls. (j–m) The PDLLA/rhBMP-2 layer mitigates early ion/alkalinity burst (j), encourages ECM deposition and Ca–P precipitation with $Mg(OH)_2$ formation (k–l), and (m) sustains osteogenesis while slowing corrosion. Adapted with permission from [59,75].

5.1.2. Angiogenesis

Mg scaffolds promote blood vessel in-growth, which is critical for large defects. In one study, a PLGA/Mg-gallate MOF scaffold loaded with stem-cell exosomes. Kang et al. [126] achieved robust vascularized bone regeneration in rat skull defects. Micro-CT and histology revealed almost complete bone healing and well-formed vessels around the Mg scaffold, whereas a similar scaffold lacking Mg showed poor bone repair. This aligns with in vitro findings that Mg^{2+} boosts endothelial proliferation and VEGF signaling.

5.2. Immunomodulation and healing environment

Mg also modulates the immune response in favor of healing. Shortly after implantation, Mg^{2+} released at the scaffold–tissue interface has been shown to encourage macrophages to adopt an M2 (anti-inflammatory) phenotype. For example, Qiao et al. [127] demonstrated that Mg^{2+} influx via the TRPM7 channel triggers kinase signaling in macrophages, leading to epigenetic changes (Histone H3 phosphorylation) that upregulate anti-inflammatory, pro-osteogenic cytokines. Gene assays around degrading Mg implants often show elevated IL-10 and reduced TNF-α compared to neutral implants, indicating faster resolution of inflammation. In short, Mg creates a pro-healing immune milieu that recruits osteoprogenitors and suppresses excessive inflammation.

5.2.1. Macrophage polarization

Mg implants increase M2-like macrophages near the defect. In animal models, the Mg group exhibits higher IL-10 (anti-inflammatory) and lower TNF-α (pro-inflammatory) gene expression compared to titanium controls, consistent with accelerated healing due to the TRPM7-mediated effect, which peaks in the early phase to aid repair [128].

5.2.2. Biphasic caution

However, if Mg remains at high concentration into the remodeling phase, it can have the opposite effect. Continued Mg exposure may over-activate NF-κB in macrophages and inhibit mineral deposition [129]. In other words, timing matters: the scaffold should degrade steadily so that Mg^{2+} levels remain within the optimal window, high enough to facilitate

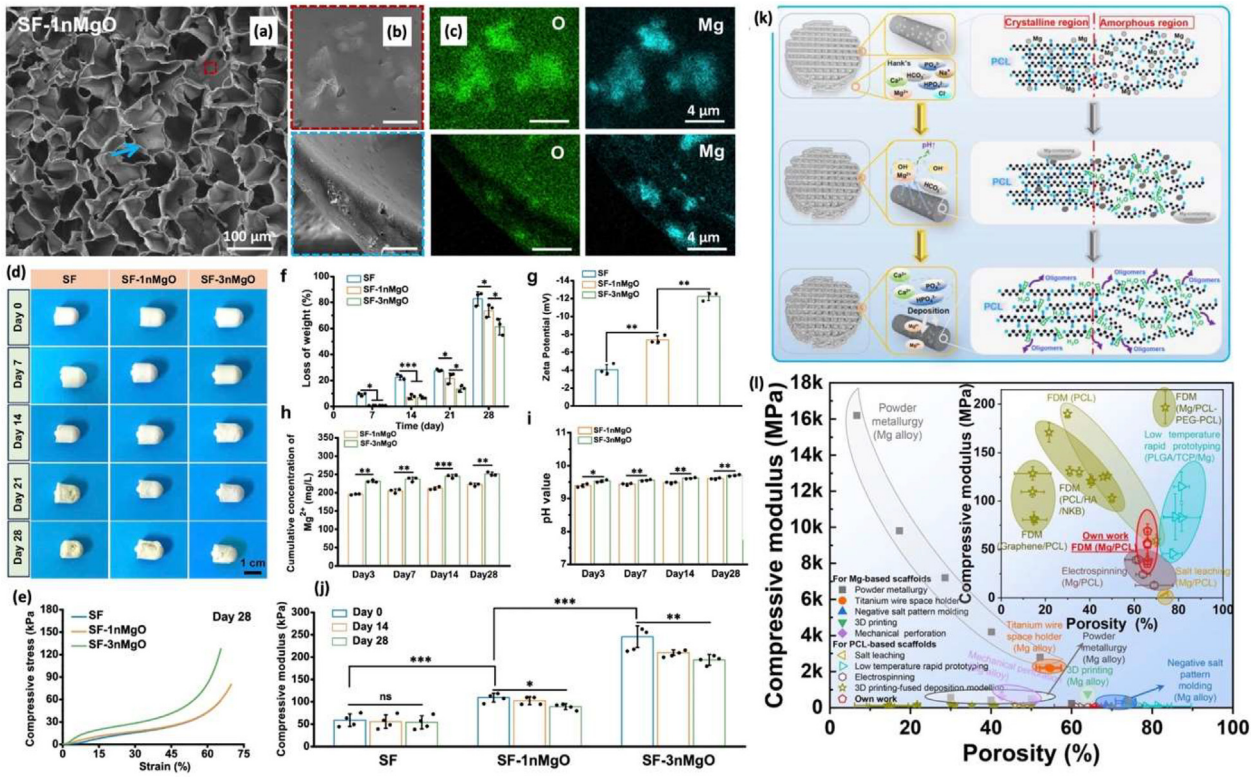


Fig. 7. (a) SEM shows a porous silk-fibroin/MgO (SF-nMgO) scaffold with interconnected walls (100 μm); (b) boxed regions reveals smooth pore walls with uniformly embedded nanoparticles; (c) homogeneous MgO dispersion along pore surfaces; (d) optical images during immersion (days 0–28) for SF, SF-1nMgO, and SF-3nMgO show progressive yet shape-retaining mass loss in MgO-containing groups; (e) compressive stress–strain indicates higher plateau stress for MgO-reinforced scaffolds versus neat SF; (f) gravimetry shows time- and loading-dependent weight loss, greatest for SF-3nMgO; (g) zeta potential becomes less negative with increasing MgO, indicating surface-charge modulation; (h) cumulative Mg^{2+} release rises with time and MgO content, (i) the surrounding pH exhibits a modest, controlled increase; (j) compressive modulus is higher and better retained in MgO-reinforced groups over 28 days than in SF (k) MgO/Mg phases releasing Mg^{2+}/OH^{-} to mildly elevate pH, neutralize acidolysis, and nucleate Ca–P deposition enhancing osteoconductivity; (l) Compressive modulus versus porosity across fabrication routes places Mg/PCL composites within a cancellous-bone-like stiffness window at high porosity. Adapted with permission from [76,113].

early healing, then low by the time bone consolidation predominates. Practically, this means designing Mg implants to resorb largely within a few months, avoiding long-lived high-Mg exposure.

6. Clinical translation challenges

Despite impressive progress in Mg scaffold research, clinical translation faces major barriers, including degradation control, mechanical reliability, manufacturing scalability, regulatory approval, and surgeon acceptance.

6.1. Control of degradation rate and hydrogen management

The greatest technical challenge is balancing degradation rate with bone healing. Too rapid corrosion leads to early loss of support and hydrogen accumulation, whereas overly slow degradation risks long-term persistence. Since weight-bearing bones require 3–6 months for consolidation [134], Mg scaffolds must retain their integrity during this period. Alloying with biocompatible elements (Zn, Ca) and coatings (fluoride, Ca–P, polymers) act as “corrosion throttles,” but coatings risk cracking and delamination. Uniform thickness and mechanical resilience of coatings remain critical [135]. Hydrogen release, although often tolerated in animals, may be problematic in humans if it forms pockets near sensitive tissues. Scaffold porosity can be engineered for gas venting, or catalytic surfaces (e.g., palladium) may convert hydrogen to water [136], though these remain experimental. Ensuring predictable degradation and safe gas dispersal is thus vital for clinical adoption.

6.2. Mechanical strength and load-bearing applications

Porous Mg matches bone’s elastic modulus but compromises strength, raising doubts for large load-bearing applications. Designs must optimize porosity for bone ingrowth while maintaining structural integrity. Modeling and topology optimization help, but unpredictable degradation can create weak spots [137]. Currently, Mg scaffolds are most promising for non-load sites or when combined with fixation hardware (plates, screws) to share mechanical loads. Future approaches include Mg composites reinforced with ceramics to enhance compressive and fatigue strength. Since bone is subject to cyclic stresses, long-term animal models under active condi-

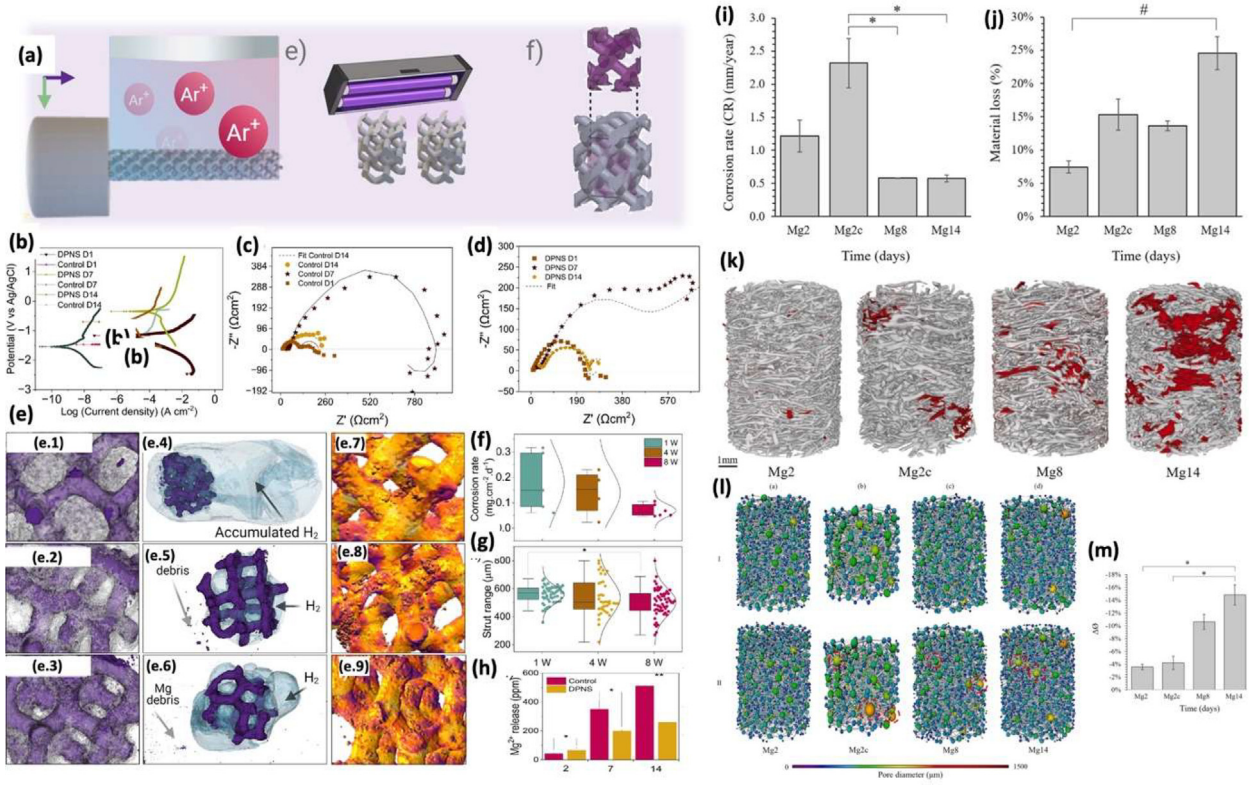


Fig. 8. (a) Ar^+ plasma–based direct plasma nano-structuring (DPNS) Mg lattice; (b) potentiodynamic polarization shows lower corrosion current density; (c–d) Nyquist plots reveal larger capacitive arcs/charge-transfer resistance after DPNS; (e) 3D micro-CT views (e.1–e.6) localized H_2 accumulation and debris around corroding struts, surface maps (e.7–e.9) highlight pitting-to-generalized corrosion transitions; (f–h) DPNS reduces total corrosion area, limits strut thinning, and lowers Mg^{2+} release over 1–8 weeks; (i–j) time-resolved corrosion rate and material loss increase with immersion duration with coated/DPNS groups; (k) longitudinal micro-CT reconstructions (red = corroded zones) illustrate progressive but more localized attack in modified lattices; (l) pore-diameter color maps (I, II) indicate architecture retention and narrower pore-size drift for treated scaffolds; (m) net porosity change ($\Delta\varphi$) confirms significantly lower volumetric loss in modified samples. Adapted with permission from [124,61].

tions are needed to assess fatigue reliability. Overall, mechanical use in minor defects is more similar to translation, while full-load-bearing applications require further development.

6.3. In vivo validation and biocompatibility

A critical step toward clinical adoption is confirming biocompatibility and functional efficacy in vivo. Preclinical animal models consistently demonstrate that Mg scaffolds promote osteoblast adhesion, angiogenesis, and new bone formation, while simultaneously degrading without causing systemic toxicity. Importantly, Mg degradation products (Mg^{2+}, hydroxides) integrate into physiological pathways, supporting bone mineralization.

However, challenges remain: rapid corrosion can cause localized alkalization and inflammatory responses, while gas pocket formation may interfere with healing. Larger animal models, particularly in weight-bearing bones, are necessary to evaluate long-term host–scaffold interactions, immunological safety, and systemic clearance. Clinical studies must provide evidence that scaffold degradation aligns with bone healing, ensuring no late-stage instability or inflammatory complications.

6.4. Manufacturing scale-up and reproducibility

Scaling Mg scaffold production from laboratory to industry requires tightly controlled, medical-grade workflows. Mg's flammability and reactivity complicate powder handling, thermal processing, and storage; however, the successful commercialization of Mg screws and plates demonstrates industrial feasibility. Uniform pore geometry remains critical because deviations directly alter corrosion kinetics and mechanical performance. Surface finishing steps may be necessary to stabilise corrosion behavior and improve coating adhesion. Sterilization also presents challenges—autoclaving accelerates corrosion, whereas gamma irradiation and ethylene oxide are suitable alternatives. Airtight packaging with desiccants is required to prevent premature degradation during storage. Achieving reproducible porosity, surface quality, and corrosion behavior across large-scale batches therefore remains a vital manufacturing hurdle.

6.5. Safety and regulatory considerations

Safety considerations extend beyond degradation by-products to include hydrogen evolution, systemic ion clearance, long-term tissue compatibility, and potential accumu-

Table 5
Key challenges for clinical translation of Mg-based scaffolds and addressing strategies.

Challenge	Description & significance	Progressive strategies
Controlling degradation rate	Mg corroding too fast, causing loss of structural support, creates severe alkalization.	• Alloy development (e.g., Mg–Zn, Mg–Ca alloys) to tune corrosion; Composite/hybrid surface coatings: MgF_2, Ca-P, polymer (PLA/PCL) significantly reduce initial corrosion burst. • Rapid Ca-P mineralization on Mg implants to provide in-situ shielding in physiological environment. • Design: lower surface area to volume ratio (thicker struts) to slow overall resorption if needed; use Mg-based bio composites. • In vivo data show conversion coatings can almost double the functional life of scaffolds.
Abrupt H_2 release	Gas pockets formation around degrading Mg. In excess, could impede healing, cause physical damage to surrounding tissue.	• Gradual degradation via coatings/dense Ca-P mineralization to keep gas release slow enough for body to absorb. • Scaffold porosity designed to vent gas (channels). • Use of catalytically active coatings (research stage) to consume H_2. • Thus far, animal studies show gas pockets are transient and not clinically significant.
Mechanical reliability	Ensuring scaffold provides sufficient mechanical support in load-bearing sites and does not fail unexpectedly.	• Optimized pore geometry having minimal strut length, optimized unit cell plays key role for strength. • Temporary co-use of fixation hardware to offload stress initially (clinically plausible strategy). • Use of higher strength Mg alloys (with rare-earth elements) for critical applications. • Fatigue testing of scaffolds in simulated body fluid to ensure safety factors.
Manufacturing scalability & quality	Reproducibly making complex Mg scaffolds at medical-grade quality; preventing oxidation during production and storage.	• Industrial SLM machines under development for Mg; some companies reports Mg devices at clinical scale implementing rigorous QA (e.g., micro-CT scanning each scaffold to check porosity, coated layer thickness measurements). • Use of vacuum and inert gas packaging with desiccants for product shelf-life. • Choosing fabrication methods amenable to scale, like press-and-sinter could be scaled with automation for simpler shapes.
Regulatory hurdles & safety data	Need to demonstrate safety (local/systemic) and efficacy in humans; lack of long-term human data.	• Leverage existing data from Mg screws and stents showing biocompatibility. • Conduct thorough preclinical studies including large animal models to satisfy safety endpoints (e.g., no organ toxicity). • Possibly start with less load-critical indications such as craniofacial defects, non-weight-bearing bones, in first human trials. • Engage regulatory bodies early to define acceptable alloys and testing standards.
Long-term outcomes & remodeling	Ensuring the bone regenerated is durable long-term; monitoring any late complications.	• Ongoing clinical studies will need to follow patients for 1–2 years to ensure pristine bone quality. • Animal studies up to 1 year (in larger models) to observe full remodeling phase with Mg scaffold fully resorbed. • Need for Adaptive strategies to flush in service implant, if scaffold is pre-fragmented during degradation rather than fully resorbed.

lation of alloying elements [138]. Regulatory bodies require evidence that Mg scaffolds exhibit predictable corrosion rates, maintain structural integrity during early healing, and do not cause pathological increases in serum Mg^{2+} [139]. Standards for biodegradable metals are still evolving, requiring tailored cytotoxicity, genotoxicity, and mechanical testing protocols [140]. Aligning manufacturing processes with GMP, ensuring batch-to-batch consistency, and demonstrating safety through large-animal models and early-phase human trials are essential steps toward regulatory approval.

In Table 5, we summarize the challenges and some strategies or progress made toward addressing them, citing relevant literature or examples.

Addressing these challenges will unlock full-scale clinical translation of Mg scaffolds by uniting programmable degradation, fatigue-robust architectures, and verified biocompatibility.

7. Conclusions

The review consolidates current understanding of Mg-based scaffolds by integrating geometric design principles, fabrication strategies, degradation science, and biological response. The evidence collectively demonstrates how architected Mg lattices can deliver mechanical competence, controlled resorption, and pro-regenerative biological performance. The key conclusions drawn from the reviewed literature are summarized below -

- Mg-based scaffolds now demonstrate cancellous-bone-range mechanics, with compressive strengths typically 4–25 MPa and elastic moduli 150–900 MPa, depending on lattice geometry and alloy system.
- Architected designs—especially TPMS gyroid/diamond lattices with 50–75% interconnected porosity and 300–

800 μm pore sizes—enable physiologic stiffness, improved permeability, and enhanced osteogenic and angiogenic responses.
- Controlled corrosion remains central: hybrid surface treatments (fluoride/Ca–P layers, LDH coatings, polymer–ceramic hybrids) reduce the initial corrosion burst, moderate pH elevation, and sustain Mg^{2+} release within pro-healing ranges, improving early stability.
- In vivo evidence consistently shows slower and more uniform degradation than static immersion tests, with transient hydrogen evolution, progressive bone ingrowth, and synchronized load transfer during the first 6–12 weeks, aligning with early healing requirements.
- Translation requires manufacturing reproducibility: additive manufacturing provides defect-conformal lattices but mandates validated build parameters, porosity QC, corrosion-fatigue characterization, and sterilization/packaging pathways that avoid premature degradation.

8. Future outlook

Despite substantial advances, several scientific and translational questions remain before Mg-based scaffolds can achieve routine clinical use. Targeted research efforts that integrate corrosion science, biomechanics, immunomodulation, and manufacturing quality will be essential to move the field forward. The following outlook highlights priority directions likely to accelerate clinical translation.

- Corrosion-mechanical co-design frameworks integrating lattice mechanics, hydrogen evolution, and resorption kinetics are essential for predictive performance.
- Geometry–biology design libraries should quantify how specific pore shapes, sizes, and TPMS architectures regulate osteogenesis, angiogenesis, and immune modulation.
- Durable multifunctional interfaces are needed to deliver stable early support, antimicrobial/anti-inflammatory cues, and controlled ion release without compromising long-term resorption.
- Clinic-ready, GMP-compliant workflows should combine patient-specific AM fabrication, lot-release imaging, and pragmatic trial designs, beginning with metaphyseal and CMF defects and progressing to load-bearing long-bone reconstruction.

Declaration of competing interest

The authors declare that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Navdeep Singh Grewal: Writing – review & editing, Conceptualization. **Mohit Verma:** Writing – original draft, Investigation. **Sanjay Kumar:** Visualization, Formal analysis. **Ahsan Riaz Khan:** Writing – review & editing, Resources. **Kamal Kumar:** Writing – review & editing, Supervision. **Neeraj Sharma:** Writing – review & editing, Validation.

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